MuleGuard

PROBLEM STATEMENT 2

the Trinetra engine · Sees the mule. Spares the look-alike. Never certifies the unseen.

Solution	MuleGuard — AI/ML Classification of Suspicious Mule Accounts (the Trinetra detection engine)
Problem Statement	PS2 — Detecting suspicious transactions & mule accounts and preventing circulation of fraudulent proceeds
Hackathon	PSB Cybersecurity, Fraud & AI Hackathon 2026 · Bank of India × IIT Hyderabad
Team	Kryptonite — Hardik Hinduja · Avinash Gehi · Sahil Deshmukh · Siddharth Dey
Dataset	Provided portal dataset (DataSet.csv) — 9,082 accounts × 3,923 features + binary target F3924

Executive summary

MuleGuard is a mule-account classifier built for the provided dataset and aligned with the national fraud-fighting stack (RBIH MuleHunter.AI, DPIP, the I4C Suspect Registry). Our central design insight is simple and decisive: the dataset is a behavioural feature matrix, so a model learns what an account does, not why it does it. A classifier can separate “behaves like a mule” from “does not” — but it cannot, from behaviour alone, tell a wilful mule from a hacked victim from a coerced victim (they look identical), and it cannot flag a dormant mule that has not yet acted. We therefore do not chase one bigger model. Trinetra wires three detection lenses in sequence — (1) Detect the behaviour accurately with a small, stable feature set; (2) Reject false positives by routing uncertainty to humans instead of freezing look-alikes; (3) Recover false negatives by refusing to certify anyone “clean” — followed by an Action Layer where an analyst plus enrichment data (account type, tenure, registry hits) decides the cause and the response.

What our EDA of the actual dataset establishes	Consequence for the solution
81 mules in 9,082 accounts — a 0.89% positive rate (~1:111 imbalance).	Accuracy is meaningless; we optimise and report PR-AUC / recall@budget, not accuracy.
3,915 numeric + 8 interpretable categorical columns (account type, opening date, tenure, region, occupation, segment).	Enables onboarding signals and a business-account whitelist — not a pure black box.
~23% of columns are quasi-dead and ~1,360 are duplicates; the matrix is ~61% zeros and 97.8% of columns have missing values.	Aggressive de-duplication + stability selection; trees with native missing-value handling.
F3912 is a probable target leak (single-feature PR-AUC 0.94; including it yields a spurious 1.000).	We quarantine F3912 and report results without it — the key guardrail against a score that fails in production.
With F3912 removed, a class-weighted gradient-boosted model reaches PR-AUC 0.908; just ~15 features reach 0.87.	A compact, defensible feature set is not a shortcut — it is the accurate choice here.

All figures above are computed by us on the provided dataset (preliminary stratified cross-validation). We deliberately report only our own results — never transferred literature numbers.

1. The problem, and why static rules fail

A money mule receives the proceeds of a fraud and rapidly forwards them, often across several banks, so the funds are split and gone before the first hop is investigated. Mule accounts are opened with forged or synthetic KYC, or rented by ordinary people for a commission, and are usually active for only a short window. Rule-based monitoring fails for three structural reasons that directly shape our design: (a) individual normality — a mule looks ordinary in isolation; the signal lives in combinations of behaviour, not any single threshold; (b) adversarial adaptation — fraudsters keep amounts under detection thresholds and concentrate abnormal activity later; and (c) volume — India processed hundreds of billions of UPI transactions in 2025, far beyond what rule lists can monitor. The deliverable is therefore a binary classifier that separates suspicious/mule accounts from legitimate ones using F1–F3923, with F3924 as the target, and — as the problem statement grades explicitly — identifies the most relevant features.

2. Dataset — detailed description

2.1 Shape, target and class balance

The provided file contains 9,082 accounts (rows) and 3,924 columns: an index column plus features F1–F3923 and the binary target F3924. The target is severely imbalanced — 81 mules (label 1) versus 9,001 legitimate accounts (label 0), a 0.89% positive rate, roughly 1 mule per 111 accounts. This single number governs the entire evaluation design (Section 5): a model that labels everyone "legitimate" would score ~99% accuracy while catching zero mules.

2.2 Feature inventory — mostly anonymised numeric, with a small interpretable core

Of the 3,923 features, 3,915 are numeric (3,876 floating-point, 39 integer) and 8 are categorical. Crucially, the dataset is not a pure black box: several categorical columns decode cleanly to meaningful account attributes, which we exploit in the Action Layer and for onboarding-time signals.

Column	Type / levels	Decoded meaning (from values)	Why it matters
F2230	4 (Sep25–Dec25)	Activity / snapshot month	Seasonality; top model-importance feature
F3886	17	Account / product type (Savings, Current, MSME Micro/Medium, Corp Adv …)	Distinguishes personal vs business → whitelist legitimate high-throughput accounts
F3888	date (back to ~2011)	Account opening date → account age	New-account-throughput & sleeper-onboarding signal
F3889	7 (e.g. L31D, L90D, L180D, G365D)	Tenure / vintage bucket	Cold-start mule entry vs aged account
F3890	4 (R, SU, M, U)	Region / locality (Rural, Semi-Urban, Metro, Urban — indicative)	Geographic risk context (Tier-1/2 patterns)
F3891	7 (salaried, selfemployed, student, agriculture, housewife …)	Occupation	Recruited-mule demographics (e.g. students)
F3892	3 (M / F / O), ~28.6% missing	Gender	Demographic context; missingness itself informative
F3893	2 (RETAIL / CORPORATE)	Customer segment	Retail vs corporate behavioural baselines

The remaining ~3,907 numeric columns are anonymised (engineered behavioural aggregates). The problem statement lists a sample of "commonly used" indices (F115, F321, F527 …); 16 of 18 are present, but their individual correlation with the target is weak (max |r| ≈ 0.06) — a clear signal that the predictive features must be found data-driven, not assumed from the hint list.

2.3 Data quality — sparse, heavily missing, highly redundant

- Missingness is pervasive: ~9.85M missing cells; 97.8% of columns contain some missing values; an average of ~1,084 missing fields per account (max 2,330). Missingness is structural (features apply only to certain account types/periods) and is itself a signal — handled natively by gradient-boosted trees and captured with missing-indicators.
- Sparsity: after treating missing as zero, ~61% of numeric cells are zero — most engineered features are inactive for most accounts.
- Redundancy & dead columns: 359 columns are constant and 885 are quasi-constant (≥99% one value) — together ~23% near-dead; an estimated ~1,360 columns are duplicates of others. Only ~3,030 columns carry usable variation.

2.4 Signal analysis and data-integrity (leakage) audit

Univariate analysis shows the linear signal is highly concentrated: only 5 features have |r| ≥ 0.20 with the target, 10 have |r| ≥ 0.10, and 122 have |r| ≥ 0.05. One feature stands out dangerously: F3912 has |r| = 0.969 and, on its own, yields PR-AUC 0.94 — a near-perfect single-feature predictor. Including it makes the whole task a spurious PR-AUC of 1.000. This is the hallmark of target leakage (a field derived from the outcome label). We therefore quarantine F3912, flag it for the organisers to verify, and report every result without it. Catching this is the difference between a headline that survives production and one that collapses the moment it meets unseen data.

2.5 What the dataset is — and is not

The file is a flat, per-account snapshot. It contains no counterparty edge list, no sender→receiver graph, and no transaction-level timeline. Three consequences follow and are reflected throughout our approach: a transaction-network graph model (GNN) cannot be trained on this file (graph is a production extension once DPIP shares counterparty signals); federated learning is not demonstrable on a single dataset (architecture only); and feature engineering on anonymised columns means principled feature selection, not domain-named feature creation.

3. Solution approach — MuleGuard Trinetra

3.1 Core principle: behaviour is not intent

Every difficult real-world case (Section 4) confuses two different things the word “mule” hides — behaviour (what the account does, which lives in the features and the classifier can learn) and cause / intent (why it does it, which is established by investigation, registry hits, KYC and account history, and is mostly not in the features). If two accounts have the same feature vector, the model must give them the same score — even if one is a criminal and the other a hacked victim. So the model’s honest job is only to sort accounts into “behaves like a mule” vs not; the cause and the response are decided in the Action Layer. Laid on a 2×2 of intent × current behaviour, the whole problem becomes legible:

	Behaves like a mule now	Behaves normal now
Intent = mule	S1 Wilful/rented · S7 Hybrid · S10 Adversarial mimic	S5 Sleeper / dormant mule
Intent = NOT mule	S2 Innocent look-alike · S3 Hacked (ATO) · S6 Coerced victim · S9 Licensed high-throughput	True legitimate (the easy case)

- The classifier can only cleave the table left-from-right (behaves-like-mule vs not). It cannot cleave top-from-bottom (intent) — so everything on the left is flagged, and which flag is a criminal vs a victim is decided downstream.
- The top-right cell (S5 sleeper) is the genuinely hard one: it sits behaviourally among legitimate accounts, so behaviour-based detection is blind to it until it activates.
- The bottom-left cell (S2, S3, S6, S9) is the false-positive battlefield — real people who look guilty. Auto-punishing them is the failure mode a bank cannot afford; the entire middle lens exists for them.

3.2 Architecture: three lenses + an Action Layer

LENS 1 · DETECT	LENS 2 · REJECT FP	LENS 3 · RECOVER FN
calibrated GBDT on a small stable feature set	hard-negative verifier + conformal abstention	PU challenger + anomaly net + never “clean”

ACTION LAYER — decides CAUSE + RESPONSE (analyst + enrichment: account type, tenure, registry, device)

Freeze + STR (wilful) · Secure customer (ATO) · Protect + warn (victim) · Whitelist (licensed) · Human REVIEW (uncertain) · Monitor (clean)

Analyst verdicts feed back as verified labels into drift detection and retraining (Section 6). Most competing systems train one model, emit one score, and force every account into mule/legitimate — the single binary that makes scenarios S2, S3, S5 and S6 unsolvable. Separating the jobs is what makes MuleGuard both accurate and safe.

3.3 Lens 1 — Detect the behaviour (accurately, with few features)

- Core model: a class-weighted gradient-boosted tree ensemble (XGBoost / LightGBM / CatBoost). On flat tabular data, gradient-boosted trees are state of the art or statistically tied for it, beating tuned deep networks while training faster and handling missing values natively (Grinsztajn et al. 2022; Shwartz-Ziv & Armon 2022).

- Imbalance — no SMOTE. We use class weights / **scale_pos_weight**. Synthetic oversampling distorts the base rate and corrupts the calibrated probabilities our freeze decision depends on; we change the metric and the loss, not the data.
- Feature selection — stability selection (Meinshausen & Bühlmann 2010; Shah & Samworth 2013): variance/duplicate filtering → correlation clustering → repeated subsampling with two base selectors (LASSO + GBDT importance), keeping only features whose selection frequency ≥ 0.6–0.8. This yields a compact (~30–60), reproducible feature set with a false-discovery guarantee — exactly the graded deliverable.
- Validated on this dataset: our EDA shows ~23% of columns are dead and ~1,360 are duplicates, and a top-15 feature model already reaches PR-AUC 0.87 vs 0.91 for the full matrix (Section 5). Fewer features is the accurate, harder-to-game, defensible choice here — not a compromise.

3.4 Lens 2 — Reject false positives (protect S2, S3, S6, S9)

- Hard-negative verifier: a second model trained on confirmed mules plus the legitimate accounts the screener most often confuses for mules, drawing the decision boundary exactly where wrongful freezes happen.
- Probability calibration: Platt scaling (small data) or isotonic regression (more data), chosen by Brier score / Expected Calibration Error on a held-out fold. A raw tree score is not a probability; we need a true one to set a cost-based cutoff.
- Conformal abstention: when evidence is insufficient the system returns “uncertain” and routes the account to a human REVIEW queue rather than forcing a label. This is what makes any “zero error in the automatic zone” statement honest.

3.5 Lens 3 — Recover false negatives (protect S4; surface S5 at activation)

- Positive-Unlabelled (PU) learning challenger (Elkan & Noto 2008; Bekker & Davis 2020): treat label-0 as unlabelled, not clean, because real fraud datasets contain undiscovered mules among the negatives. An account the PU model flags despite a 0 → URGENT REVIEW. (Activated only after a label-trust audit; if F3924 is clean ground truth, the calibrated GBDT alone suffices.)
- Unsupervised anomaly net (Isolation Forest): a label-free second opinion that can catch patterns the supervised model never saw, and the “too-perfectly-normal” adversarial mimic.
- Never certify “clean”. The benign state is “not currently flagged + monitored”, so a missed mule is never stamped innocent.

3.6 The Action Layer — where cause and response are decided

The model emits a behavioural verdict plus a reason code (top SHAP-driving feature clusters mapped to known mule typologies — fan-in, rapid pass-through, dormancy-then-burst, structuring, new-account throughput). Because several dataset columns are interpretable (account type, tenure, region), the Action Layer can resolve the cause and choose a proportionate response:

Reason-code pattern	Likely cause	Response
Fan-in + pass-through + near-zero retention, on an established account	Wilful / rented mule (S1)	Freeze → counterfactual evidence packet → STR/SAR → DPIP/I4C alert
Sudden break from the account's own history; device change	Hacked / ATO (S3)	Secure the customer, step-up auth, contain loss — not prosecute
Mule-like flow + scam pattern; victim demographics	Coerced victim (S6)	Protect + warn, recover funds, victim support
Fan-in/out on a Current / MSME / Corp account (F3886, F3893)	Licensed / business (S9)	Whitelist, stop re-flagging
Borderline / model disagreement / PU flag / out-of-distribution	Unknown (S2, S5, S7)	Ranked human REVIEW; verdict feeds back as a label
Low calibrated probability + high agreement + low anomaly	True legitimate	Monitor, no action

Counterfactual evidence packet (per confirmed mule): the minimal feature change that would clear the flag, mapped to a typology — e.g. “flagged: rapid pass-through + fan-in from many small senders; if the rapid-cycle ratio fell from 0.94 to the legitimate-cohort 0.22, conviction would drop to 0.11.” This is the artefact a compliance officer can file and a regulator will accept.

4. Robustness to real-world account scenarios

We stress-tested the design against ten ways an account can confuse a mule detector — including the hardest edge cases. Each is mapped to the lens that defends it; the honest limits are stated in Section 8.

#	Scenario	Concrete example	How MuleGuard handles it
S1	Wilful / rented mule	Account sold or rented to move dirty money	Detected by Lens 1 (genuinely mule-like behaviour) → freeze + STR

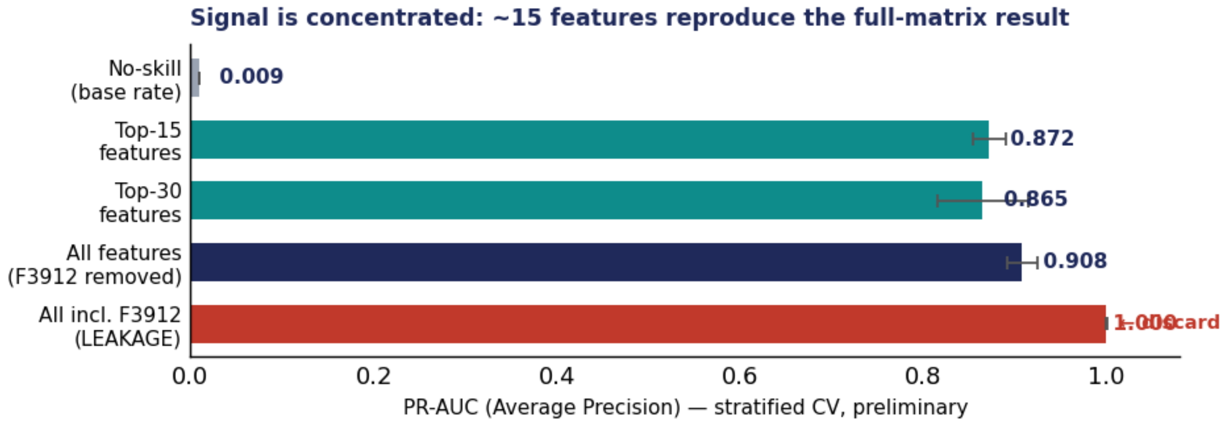
#	Scenario	Concrete example	How MuleGuard handles it
S2	Innocent look-alike (false-positive risk)	Kirana shop aggregating many small UPI inflows; a car-sale; wedding/festival money; group-trip collection	Lens 2 hard-negative verifier + conformal abstention → REVIEW, never an auto-freeze
S3	Hacked / account takeover	A genuine customer's account compromised and used as a mule	Flagged, but a sudden break from the account's own history routes it as “secure the customer”, not prosecute
S4	Missed mule (false negative — most costly)	A real mule scored legitimate; money already gone	Lens 3: PU challenger + anomaly net + never certify “clean”; audited miss-rate with a confidence bound
S5	Sleeper / dormant mule	Opened with synthetic KYC, kept normal-looking to “age” before activation	Onboarding screen via age/tenure features (present in data) + fast detection at activation; not claimed pre-activation
S6	Coerced / unwitting victim	Romance scam, fake “payment-processing job”, “move funds to a safe account”	Same flag, humane action: protect + warn the customer, recover funds
S7	Hybrid (real + mule) account	A genuine shop that also launders on the side; aggregate features dilute the signal	Within-account heterogeneity features; otherwise routed to REVIEW by design
S8	Coordinated ring, each account individually weak	No single account crosses threshold, but the network is a ring	Members crossing threshold are caught; the rest are the explicit graph-layer roadmap (no edges in this file)
S9	Licensed high-throughput entity	Payment aggregator, remittance agent, corporate account	Account-type/segment columns → whitelist; stop re-flagging legitimate businesses
S10	Adversarial mimic	A mule deliberately shaped to look like a legitimate cohort	Relational ratio features that survive single-amount gaming + anomaly net + hidden thresholds + drift detection

5. Evaluation methodology and honest claims

Under a 0.89% base rate, accuracy and ROC-AUC are misleading. We report the precision-recall family, which is the honest view for rare-event detection (Saito & Rehmsmeier 2015), and translate it into operational terms a bank uses.

Metric	What it answers	Why a bank cares
PR-AUC (Average Precision)	Skill at finding rare mules across thresholds	The honest headline under imbalance
Recall @ alert budget	Of the top-N an analyst can review per day, how many mules are caught	Maps to staffed review capacity
Precision in the FREEZE tier	Of auto-frozen accounts, the fraction truly mule	Bounds wrongful-freeze (S2, S3, S6, S9)
Recall @ fixed low FPR	Mules caught while the innocent-freeze rate stays tiny	The real operating constraint
Coverage + selective error	% auto-decided vs routed; error only among auto-decisions	The honest “zero-error” metric
Brier score / ECE	Whether probabilities are trustworthy	The score drives a freeze, not just a flag
Rs recovered + analyst-hours	Output in money and labour	The bottom-line business case

5.1 Preliminary results on the provided dataset



Class-weighted LightGBM, no SMOTE, stratified cross-validation (preliminary). With the leakage feature F3912 removed, PR-AUC = 0.908 ± 0.016 (ROC-AUC 0.998) against a no-skill PR-AUC of 0.009. A top-15 feature model reaches 0.872 and a top-30 model 0.865 — i.e. roughly fifteen features reproduce the full-matrix result, validating our compact-feature strategy. Including F3912 produces a spurious PR-AUC of 1.000, which we discard.

5.2 Validation hygiene and the claims we make

- Stratified k-fold with all preprocessing (selection, scaling, calibration) fit inside each fold; a locked test set touched once; the original base rate preserved in validation and test. We never evaluate on a balanced/oversampled test set.
- We do not claim “100% accuracy” or “zero false positives”. The defensible statement is: “no errors were observed in the automatic decision zone of the locked test set at the reported coverage; uncertain cases were routed to review”, accompanied by a 95% confidence bound — zero errors in N cases implies an upper bound of about 3/N.
- Inference is deterministic and auditable (versioned models, calibrated uncertainty, human-in-the-loop on every freeze) — but determinism does not mean correctness, so we rely on calibration and review, not absolute guarantees.

6. Adapting as fraudsters evolve (concept drift)

Static detectors decay as adversaries change behaviour (Gama et al. 2014). MuleGuard monitors the prediction-score distribution with ADWIN drift detection (Bifet & Gavaldà 2007), triggers champion–challenger evaluation and sliding-window retraining, and feeds confirmed analyst verdicts back as fresh labels. We monitor both real drift (the feature→label relationship shifts) and virtual drift (only the input distribution shifts). Relational ratio features (rapid-cycle, fan-in concentration, retention) are preferred precisely because they survive single-amount gaming, and detection thresholds are never exposed for fraudsters to learn.

7. Ecosystem & regulatory alignment

MuleGuard is designed to be aligned with and complementary to the national stack, not a replacement for it:

- RBIH MuleHunter.AI — in-house AI/ML mule detection (RBIH states 85%+ accuracy), studying 19 mule-behaviour patterns, launched December 2024 and live in 26 banks as of March 2026, designed to freeze suspicious transactions before withdrawal. MuleGuard adds calibrated, reason-coded, abstaining decisions and counterfactual evidence packets that fit this ecosystem.
- MHA directive — all financial institutions to integrate with MuleHunter by December 2026, making a MuleHunter-aligned design timely.
- DPIP (RBIH Digital Payments Intelligence Platform) — the real-time cross-bank signal-sharing layer where the problem statement's “real-time feeds / cross-channel data” live in production, and the source of the counterparty edges for the future graph layer (S8).
- I4C–RBIH MoU (May 2026) — the Suspect Registry feeds AI detection; in MuleGuard a confirmed-registry hit short-circuits the score (override layer) and helps resolve S6/S9.
- RBI KYC Direction on money mules — the diligence, monitoring and suspicious-transaction-reporting obligations MuleGuard helps satisfy.

8. Scope, assumptions and honest limits

Stating the boundary is a feature, not a weakness — it is how a bank tells a production-ready system from a demo:

- Behaviour is not intent. From this data the model cannot distinguish a wilful mule from a hacked or coerced victim; cause is decided by analysts and enrichment data (S1 vs S3 vs S6).

- Dormant mules are largely undetectable from behaviour until they act; we screen at onboarding (age/tenure features are present) and detect quickly at activation, but we do not claim pre-activation detection (S5).
- Coordinated rings are invisible on a flat file; individually-weak members need the counterparty graph — a production roadmap, not this model (S8).
- No zero-miss, no zero-false-positive. We report bounded, audited rates with confidence bounds and route uncertainty to humans.
- Literature metrics do not transfer. Numbers from credit-card, e-commerce or synthetic datasets are methodological evidence only; we report exclusively our own cross-validation on the provided dataset.
- A model is not a system. Without the Action Layer (registry, KYC, device, human review, whitelist), detection alone turns victims into defendants and licensed entities into perpetual false alarms.

9. Implementation stack & build plan

Stage	Deliverable
1. EDA & hygiene	Profile 3,923 columns; drop quasi-constant/duplicate; confirm the 0.89% class ratio; quarantine F3912; baseline PR-AUC
2. Feature selection	Correlation-cluster → stability selection (LASSO + GBDT importance, freq ≥ 0.6–0.8) → ~30–60 features + selection-frequency table; ablation-confirm PR-AUC
3. Lens 1 (Detect)	Class-weighted CatBoost/LightGBM/XGBoost; no SMOTE; low-threshold high-recall screener
4. Lens 2 (Reject FP)	Hard-negative mining loop; Platt/isotonic calibration (Brier/ECE); conformal abstention → REVIEW
5. Lens 3 (Recover FN)	Label-trust audit → PU challenger if noisy; Isolation Forest anomaly net; never certify clean
6. Action Layer	SHAP-clustered reason codes + counterfactual packet; route by cause; account-type/registry enrichment & whitelist
7. Evaluation	Nested CV (preprocessing inside the fold), original base rate; PR-AUC, recall@budget, freeze-tier precision, coverage, selective error, Rs-recovered
8. Robustness	ADWIN/DDM drift + sliding-window retraining + analyst-feedback loop

Stack: Python · pandas/Polars · scikit-learn (LASSO, permutation importance, stability-selection loop) · XGBoost / LightGBM / CatBoost · PU-learning (Elkan–Noto) · Isolation Forest · CalibratedClassifierCV (isotonic + Platt) · MAPIE (conformal) · SHAP + Anchor · River (ADWIN/DDM) · FastAPI + Docker (versioned model bundle) · React + Recharts analyst dashboard · MLflow / SQLite model registry.

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MuleGuard · Trinetra — Team Kryptonite · PS2 · PSB Cybersecurity, Fraud & AI Hackathon 2026 (Bank of India × IIT Hyderabad). All dataset figures and model results computed by the team on the provided dataset; performance reported only from our own cross-validation.